



2014 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Tuesday 4th March 2014

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 130 boys

Examiner
LRP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1

What is the period of the function $f(x) = 3 \tan 2x$?

- (A) 4π
- (B) 2π
- (C) π
- (D) $\frac{\pi}{2}$

QUESTION TWO

1

Which of the following is the derivative of $\sin^{-1} 2x$?

- (A) $\frac{2}{\sqrt{1-4x^2}}$
- (B) $\frac{-2}{\sqrt{1-4x^2}}$
- (C) $\frac{1}{2\sqrt{1-4x^2}}$
- (D) $\frac{-1}{2\sqrt{1-4x^2}}$

QUESTION THREE

1

What is the general solution of $2 \cos x + 1 = 0$?

- (A) $x = 2n\pi \pm \frac{\pi}{3}, n$ is an integer
- (B) $x = 2n\pi \pm \frac{2\pi}{3}, n$ is an integer
- (C) $x = 2n\pi \pm \frac{\pi}{6}, n$ is an integer
- (D) $x = 2n\pi \pm \frac{5\pi}{6}, n$ is an integer

Exam continues next page ...

QUESTION FOUR

1

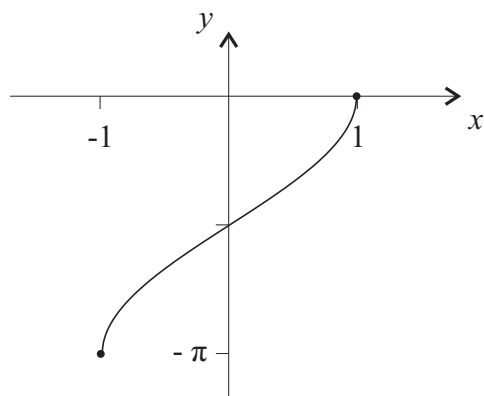
What is the range of the function $y = 2 \cos^{-1} 3x$?

- (A) $-2 \leq y \leq 2$
- (B) $0 \leq y \leq \frac{\pi}{3}$
- (C) $-\pi \leq y \leq \pi$
- (D) $0 \leq y \leq 2\pi$

QUESTION FIVE

1

Which function does the graph below represent?



- (A) $y = \cos^{-1} x - \pi$
- (B) $y = \cos^{-1}(-x)$
- (C) $y = \sin^{-1} x - \frac{\pi}{2}$
- (D) $y = \sin^{-1} x - \pi$

QUESTION SIX

1

The angle θ satisfies $\cos \theta = \frac{4}{5}$ and $-\frac{\pi}{2} < \theta < 0$. What is the value of $\sin 2\theta$?

- (A) $\frac{24}{25}$
- (B) $-\frac{24}{25}$
- (C) $\frac{7}{25}$
- (D) $-\frac{7}{25}$

QUESTION SEVEN

1

Which of the following represents the inverse function of $f(x) = \frac{2}{x-3} - 6$?

(A) $f^{-1}(x) = 6 - \frac{2}{x-3}$

(B) $f^{-1}(x) = \frac{2}{x+6} - 3$

(C) $f^{-1}(x) = \frac{2}{x} - 3$

(D) $f^{-1}(x) = \frac{2}{x+6} + 3$

QUESTION EIGHT

1

How many solutions are there to the equation $\sin 3x = 0$, where $0 \leq x \leq 2\pi$?

(A) 1

(B) 5

(C) 7

(D) 9

QUESTION NINE

1

The volume, $V \text{ cm}^3$, of water in a container is given by $V = \frac{1}{3}\pi h^3$, where $h \text{ cm}$ is the depth of water in the container at time t minutes. Water is draining from the container at a constant rate of $100 \text{ cm}^3/\text{min}$. What is the rate of decrease of h , in cm/min , when $h = 5$?

(A) $\frac{\pi}{4}$

(B) $\frac{4}{\pi}$

(C) 25π

(D) 2500π

Exam continues next page ...

QUESTION TEN

1

A trigonometric function has the properties $f(\pi - x) = -f(x)$ and $f(\pi - x) = -f(-x)$ for all real values of x . Which of the following is a possible equation of this function?

- (A) $f(x) = \sin x$
- (B) $f(x) = \cos x$
- (C) $f(x) = \tan x$
- (D) $f(x) = \operatorname{cosec} x$

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.	Marks
(a) Find the acute angle between the lines $y = 2x + 3$ and $y = -3x + 1$.	2
(b) Write down the exact value of:	
(i) $\sin \frac{5\pi}{3}$	1
(ii) $\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right)$	1
(c) Differentiate:	
(i) $\sin \frac{x}{2}$	1
(ii) $(\cos^{-1} x)^2$	2
(iii) $x \tan 2x$	2
(d) Find:	
(i) $\int \cos (2x + 1) dx$	1
(ii) $\int \frac{3}{9 + x^2} dx$	1
(e) (i) Show that $\tan x = \frac{\sin 2x}{1 + \cos 2x}$.	2
(ii) Hence evaluate $\tan \frac{\pi}{8}$ in simplest exact form.	2

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) Find the exact value of $\tan \left(2 \cos^{-1} \frac{3}{4} \right)$. **2**

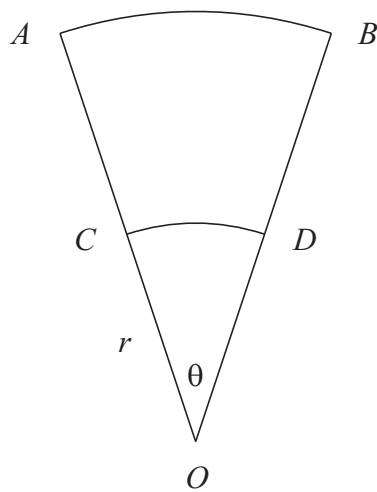
(b) The two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are on the parabola $x^2 = 4ay$.
The equation of the tangent to $x^2 = 4ay$ at an arbitrary point $(2at, at^2)$ is $y = tx - at^2$.
(Do NOT prove this.)

(i) Show that the tangents at the points P and Q meet at $R(a(p+q), apq)$. **2**

(ii) Show that R lies on the directrix if the tangents at P and Q are perpendicular. **2**

(c) Solve $\cos 2\theta + 3 \sin \theta - 2 = 0$ for $0 \leq \theta \leq 2\pi$. **3**

(d)



The diagram above shows a sector OAB and an arc CD , both with centre O . The figure is formed by pieces of wire. The area of sector OAB is four times the area of sector OCD .

The length of OC is r cm and $\angle AOB$ is θ .

(i) Show that $AC = r$. **2**

(ii) If the total length of wire forming the figure is 48 cm, show that $\theta = \frac{48 - 4r}{3r}$. **1**

(iii) Hence find the value of r that maximises the area of sector OCD . **3**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

(a) Use the substitution $t = \tan \frac{\theta}{2}$ to show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$. **2**

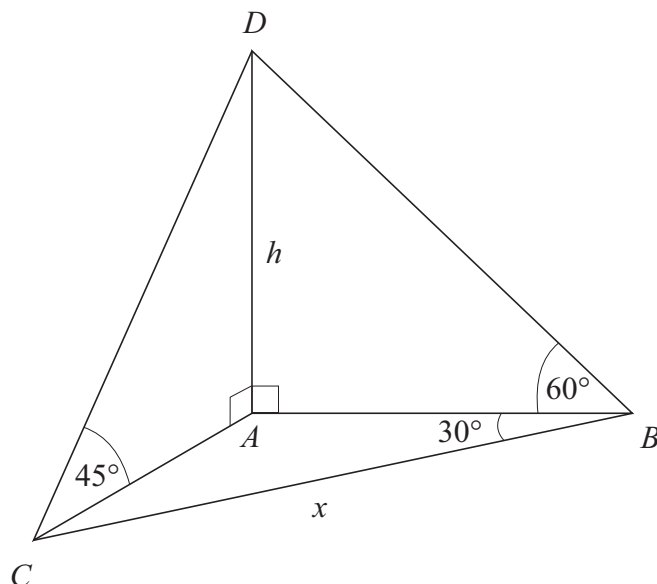
(b) (i) Express $2 \sin x + 4 \cos x$ in the form $R \sin(x + \alpha)$, where $R > 0$ and $0 \leq \alpha < 360^\circ$. Give α correct to the nearest minute. **2**

(ii) Hence, or otherwise, solve the equation $2 \sin x + 4 \cos x = 3$, for $0 \leq x \leq 360^\circ$. Give your solutions correct to the nearest minute. **3**

(c) Use mathematical induction to prove that for all integers $n \geq 1$, **3**

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}.$$

(d)



In the diagram above, $ABCD$ is a triangular pyramid with base ABC and perpendicular height $AD = h$.

It is given that $\angle ABC = 30^\circ$, $\angle ACD = 45^\circ$ and $\angle ABD = 60^\circ$. Let BC be x .

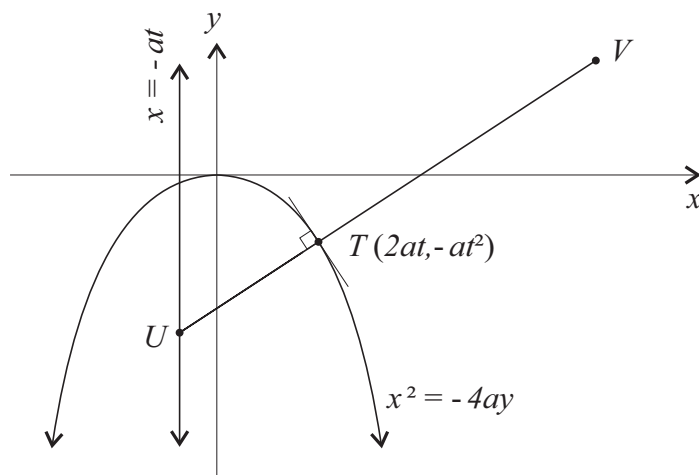
(i) Show that $AB = \frac{h}{\sqrt{3}}$ and $AC = h$. **1**

(ii) Use the cosine rule to show that $2h^2 + 3xh - 3x^2 = 0$. **2**

(iii) Hence show that $\frac{h}{x} = \frac{\sqrt{33} - 3}{4}$. **2**

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Consider the function $y = 3 \cos^{-1}(x - 1)$.(i) Draw a neat sketch of $y = 3 \cos^{-1}(x - 1)$, showing the intercepts with the axes. 2(ii) Find the exact area enclosed by $y = 3 \cos^{-1}(x - 1)$ and the coordinate axes. 2(b) A function is defined by $f(x) = \sin(x - \frac{\pi}{6})$, where $-a \leq x \leq a$. 2If the inverse of $f(x)$ is a function, what is the maximum possible value of a ?

(c)



The point $T(2at, -at^2)$ is a point on the parabola $x^2 = -4ay$. The normal at T meets the line $x = -at$ at point U . Point V lies on the normal and divides TU externally in the ratio $2 : 3$.

(i) Show that the equation of the normal to the parabola at T is $x - ty = 2at + at^3$. 2(ii) Find the coordinates of U in terms of a and t . 2(iii) Find the coordinates of V in terms of a and t . 2(iv) The locus of V is another parabola. Find the coordinates of its focus. 3

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

2014 HALF-YEARLY → SOLUTIONS

Form VI - Mathematics Extension 1

Q1	D
Q2	A
Q3	B
Q4	D
Q5	C
Q6	B
Q7	D
Q8	C
Q9	B
Q10	B

Multiple Choice Working

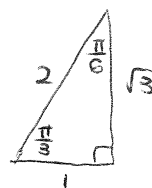
Q1. $f(x) = 3 \tan 2x$

$$\text{Period} = \frac{\pi}{2}$$

$$\begin{aligned} \text{Q2. } \frac{d}{dx} (\sin^{-1} 2x) &= \frac{1}{\sqrt{1-(2x)^2}} \times 2 \\ &= \frac{2}{\sqrt{1-4x^2}} \end{aligned}$$

Q3. $\cos x = -\frac{1}{2}$

$$\begin{aligned} \therefore x &= 2n\pi \pm \cos^{-1}\left(-\frac{1}{2}\right) \\ &= 2n\pi \pm \frac{2\pi}{3} \end{aligned}$$



$$\begin{aligned} \alpha &= \pi - \frac{\pi}{3} \\ &= \frac{2\pi}{3} \end{aligned}$$

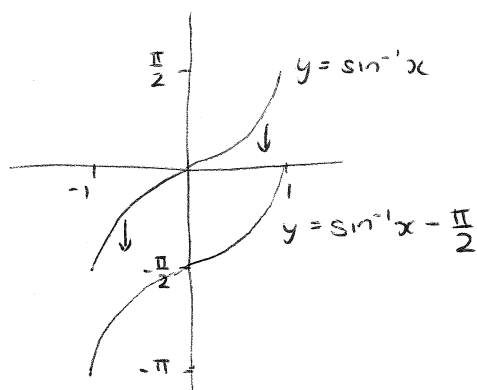
Q4. $y = 2\cos^{-1} 3x$

$$0 \leq \cos^{-1} 3x \leq \pi$$

$$2 \times 0 \leq 2\cos^{-1} 3x \leq 2\pi$$

$$\therefore 0 \leq y \leq 2\pi$$

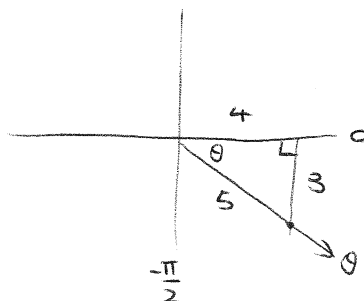
Q5.



Q6. $\sin 2\theta$
 $= 2\sin\theta\cos\theta$

$$= 2 \times -\frac{3}{5} \times \frac{4}{5}$$

$$= -\frac{24}{25}$$

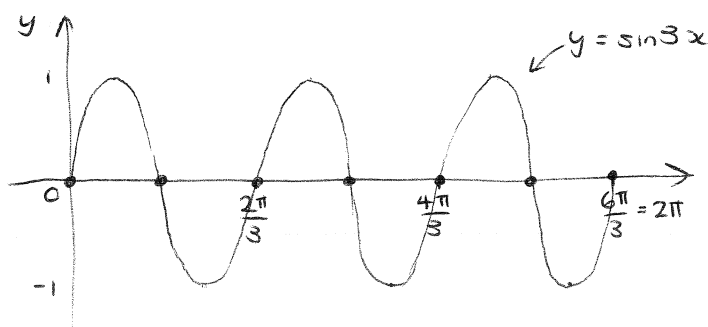


Q7. $x = \frac{2}{y-3} - 6$

$$y-3 = \frac{2}{x+6}$$

$$\therefore y = 3 + \frac{2}{x+6}$$

Q8.



Period = $\frac{2\pi}{3}$

Q9. $\frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt}$

$$100 = \pi h^2 \times \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{100}{\pi h^2}$$

when $h = 5$, $\frac{dh}{dt} = \frac{100}{\pi \times 25}$

$$= \frac{4}{\pi} \text{ cm/min}$$

Q10. $f(x) = \cos x$ is positive in the 1st & 4th quadrants
and negative in the 2nd quadrant

$$\therefore f(\pi - \theta) = -f(\theta)$$

$$\neq f(\pi - \theta) = -f(-\theta)$$

also $f(x) = f(-x) \rightarrow$ even function

$\therefore f(x) = \cos x$ is the only option.

QUESTION 11:

a) $m_1 = 2$
 $m_2 = -3$

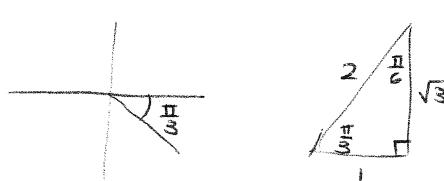
$$\alpha = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \tan^{-1} \left| \frac{2 - (-3)}{1 + 2 \times (-3)} \right| \quad \checkmark$$

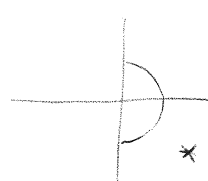
$$= \tan^{-1} 1$$

$$= 45^\circ \text{ (or } \frac{\pi}{4} \text{)} \quad \checkmark$$

b) i) $\sin \frac{5\pi}{3} = -\sin \frac{\pi}{3}$

$$= -\frac{\sqrt{3}}{2} \quad \checkmark$$


ii) $\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) = -\frac{\pi}{6}$
 (or -30°) $\checkmark$



c) i) $\frac{d}{dx} \left(\sin \frac{x}{2} \right) = \cos \frac{x}{2} \times \frac{1}{2}$

$$= \frac{1}{2} \cos \frac{x}{2} \quad \checkmark$$

ii) $\frac{d}{dx} \left((\cos^{-1} x)^2 \right) = 2 \cos^{-1} x \times \frac{1}{\sqrt{1-x^2}} \quad \checkmark$

$$= -\frac{2 \cos^{-1} x}{\sqrt{1-x^2}} \quad \checkmark$$

iii) $\frac{d}{dx} (x \tan 2x)$

$$= \tan 2x \times 1 + x \times 2 \sec^2 2x \quad \checkmark$$

$$= \tan 2x + 2x \sec^2 2x \quad \checkmark$$

Let $u = x$ $v = \tan 2x$
 $u' = 1$ $v' = 2 \sec^2 2x$

$$d) i) \int \cos(2x+1) dx = \frac{1}{2} \sin(2x+1) + C \quad \checkmark$$

$$\begin{aligned} ii) \int \frac{3}{9+x^2} dx &= \int \frac{3}{3^2+x^2} dx \\ &= \frac{3}{3} \tan^{-1} \frac{x}{3} + C \\ &= \tan^{-1} \frac{x}{3} + C \quad \checkmark \end{aligned}$$

$$\begin{aligned} e) i) \text{ RHS} &= \frac{\sin 2x}{1 + \cos 2x} \\ &= \frac{2 \sin x \cos x}{2 \cos^2 x} \quad \checkmark (\text{for either one}) \\ &= \frac{\sin x}{\cos x} \\ &= \tan x \\ &= \text{LHS as required.} \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \checkmark$$

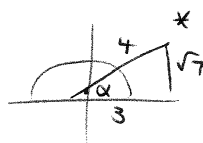
$$\begin{aligned} ii) \tan \frac{\pi}{8} &= \frac{\sin(2 \times \frac{\pi}{8})}{1 + \cos(2 \times \frac{\pi}{8})} \\ &= \frac{\sin \frac{\pi}{4}}{1 + \cos \frac{\pi}{4}} \quad \checkmark \\ &= \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \left(\times \frac{\sqrt{2}}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{2} + 1} \quad (\text{or } \sqrt{2} - 1) \quad \checkmark (\text{either}) \end{aligned}$$

QUESTION 12:

a) $\tan(2\cos^{-1}\frac{3}{4})$

Let $\cos^{-1}\frac{3}{4} = \alpha$

$\therefore \cos \alpha = \frac{3}{4} \quad 0 \leq \alpha \leq \pi$



$\tan \alpha = \frac{\sqrt{7}}{3}$ ✓

$\tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$

$= \frac{2 \times \frac{\sqrt{7}}{3}}{1 - \frac{7}{9}}$

$= 3\sqrt{7}$ ✓

b) i) Tangent at P: $y = px - ap^2$ ①

" Q: $y = qx - aq^2$ ②

① - ②: $(p - q)x - a(p^2 - q^2) = 0$ ✓

$$x = \frac{a(p+q)(p-q)}{(p-q)} \quad [p \neq q \text{ assuming } P, Q \text{ distinct}]$$

$$= a(p+q)$$

Sub into ①:

$$y = p(ap + aq) - ap^2$$

$$= apq$$

$\therefore R(a(p+q), apq)$

ii) Gradient of tangent at P: $m_p = p$

" Q: $m_q = q$

If perpendicular, then $m_p \times m_q = -1$

$\therefore pq = -1$ ✓

$$y\text{-coord of } R = apq$$

$$= ax - 1$$

$$= -a$$

$\therefore R$ lies on the directrix $y = -a$

$$c) \cos 2\theta + 3\sin\theta - 2 = 0 \quad 0 \leq \theta \leq 2\pi$$

$$1 - 2\sin^2\theta + 3\sin\theta - 2 = 0$$

$$2\sin^2\theta - 3\sin\theta + 1 = 0$$

$$(2\sin\theta - 1)(\sin\theta - 1) = 0$$

$$\therefore \sin\theta = \frac{1}{2}$$

$$\text{or } \sin\theta = 1$$

$$\therefore \theta = \frac{\pi}{2}$$

S	A	Related = $\frac{\pi}{6}$
T	C	

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\therefore \theta = \frac{\pi}{6}, \frac{\pi}{2} \text{ or } \frac{5\pi}{6}$$

$$d) i) \frac{1}{2} \times OA^2 \times \theta = 4 \times \frac{1}{2} \times OC^2 \times \theta$$

$$OA^2 = 4r^2$$

$$\therefore OA = 2r \quad (\text{since } OA > 0)$$

$$AC = OA - OC$$

$$= 2r - r$$

$$= r$$

$$ii) 48 = 2r + 2r + 2r\theta + r\theta$$

$$= 4r + 3r\theta$$

$$\therefore \theta = \frac{48 - 4r}{3r} \text{ as required}$$

$$\text{iii) } A_{\text{ocD}} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times r^2 \times \frac{48 - 4r}{3r} \quad \checkmark$$

$$= \frac{48r}{6} - \frac{4r^2}{6}$$

$$= 8r - \frac{2}{3} r^2$$

$$\frac{dA}{dr} = 8 - \frac{4}{3} r \quad \checkmark$$

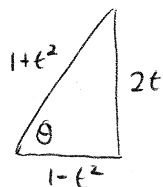
$$= 0 \quad \text{when } r = 6$$

$$\frac{d^2A}{dr^2} = -\frac{4}{3}$$

$$< 0 \quad \checkmark \quad \therefore \text{max occurs when } r = 6 \quad \checkmark \quad \begin{matrix} \text{(must} \\ \text{justify max)} \end{matrix}$$

QUESTION 13:

a)



$$\text{LHS} = \operatorname{cosec} \theta + \cot \theta$$

$$= \frac{1}{\sin \theta} + \frac{1}{\tan \theta}$$

$$= \frac{1+t^2}{2t} + \frac{1-t^2}{2t} \quad \checkmark \text{ (for either)}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \frac{1}{\tan \frac{\theta}{2}}$$

$$= \cot \frac{\theta}{2}$$

$$= \text{RHS as required.}$$

$$\text{b) i) } 2 \sin x + 4 \cos x = R \sin(x + \alpha)$$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$R \cos \alpha = 2 \quad \text{①}$$

$$R \sin \alpha = 4 \quad \text{②}$$

Squaring & adding:

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 2^2 + 4^2$$

$$= 20$$

$$\therefore R = 2\sqrt{5} \quad (R > 0) \quad \checkmark$$

$$\therefore \cos \alpha = \frac{2}{2\sqrt{5}}$$

$$= \frac{1}{\sqrt{5}}$$

$$\sin \alpha = \frac{4}{2\sqrt{5}}$$

$$= \frac{2}{\sqrt{5}}$$

S	A ^x
T	C

$$\alpha = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

$$= 63.4349\dots$$

$$= 63^\circ 26' \text{ (to nearest min)}$$

$$\therefore 2 \sin x + 4 \cos x = 2\sqrt{5} \sin(x + 63^\circ 26')$$

↑
(to nearest minute)

ii) $2\sqrt{5} \sin(x + 63^\circ 26') = 3$

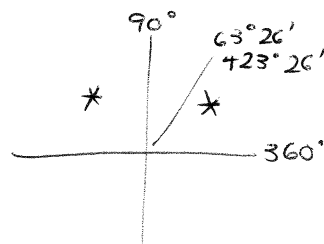
$$\sin(x + 63^\circ 26') = \frac{3}{2\sqrt{5}}$$

$$\text{Related } \angle = \sin^{-1}\left(\frac{3}{2\sqrt{5}}\right)$$

$$= 42.1304...^\circ \quad \checkmark$$

$$0^\circ \leq x \leq 360^\circ$$

$$63^\circ 26' \leq x + 63^\circ 26' \leq 423^\circ 26'$$



$$x + 63^\circ 26' = 180^\circ - 42.1304...^\circ \quad \text{OR} \quad 360^\circ + 42.1304...^\circ$$

$$= 137.8695...^\circ \quad \quad \quad = 402.1304...^\circ \quad \checkmark$$

$$\therefore x = 74^\circ 26' \quad \text{OR} \quad 338^\circ 42' \quad \checkmark$$

c) Test first case:

When $n=1$:

$$\text{LHS} = \frac{1}{2}$$

$$\text{RHS} = 2 - \frac{1+2}{2}$$

$$= \frac{1}{2}$$

$$= \text{LHS} \quad \therefore \text{true for } n=1 \quad \checkmark$$

Assume true for $n=k$:

$$\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^k} = 2 - \frac{k+2}{2^k}$$

Prove true for $n=k+1$:

$$\left[\text{Required to prove } \Sigma = 2 - \frac{k+3}{2^{k+1}} \right]$$

$$\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^k} + \frac{k+1}{2^{k+1}} = 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}} \quad \checkmark$$

$$= 2 - \frac{2(k+2)}{2(2^k)} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \left(\frac{2k+4}{2^{k+1}} - \frac{k+1}{2^{k+1}} \right)$$

$$= 2 - \frac{k+3}{2^{k+1}} \quad \text{as required.} \quad \checkmark$$

The result now follows for all integers ≥ 1 by the Principle of Mathematical Induction.

$$d) i) \tan 60^\circ = \frac{h}{AB}$$

$$\therefore AB = \frac{h}{\tan 60^\circ}$$

$$= \frac{h}{\sqrt{3}}$$

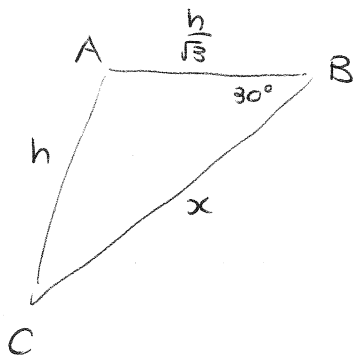
$$\tan 45^\circ = \frac{h}{AC}$$

$$\therefore AC = \frac{h}{\tan 45^\circ}$$

$$= \frac{h}{1}$$

$$= h$$

ii)



$$h^2 = x^2 + \left(\frac{h}{\sqrt{3}}\right)^2 - 2 \times x \times \frac{h}{\sqrt{3}} \times \cos 30^\circ$$

$$= x^2 + \frac{h^2}{3} - \frac{2xh}{\sqrt{3}} \times \frac{\sqrt{3}}{2}$$

$$3h^2 = 3x^2 + h^2 - 3xh$$

$$\therefore 2h^2 + 3xh - 3x^2 = 0 \text{ as required.}$$

iii) METHOD 1:

$$2h^2 + 3xh - 3x^2 = 0$$

*

$$a = 2$$

$$b = 3x$$

$$c = -3x^2$$

$$h = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3x \pm \sqrt{(3x)^2 - 4 \times 2 \times -3x^2}}{2 \times 2}$$

$$= \frac{-3x \pm \sqrt{33x^2}}{4}$$

$$= \left(\frac{-3 \pm \sqrt{33}}{4} \right) x$$

BUT $h > 0 \quad \therefore h = \left(\frac{-3 + \sqrt{33}}{4} \right) x$

$$\neq \frac{h}{x} = \frac{-3 + \sqrt{33}}{4}$$

✓

✓ (must justify positive answer only)

METHOD 2:

*

divide throughout by x^2 :

$$2 \frac{h^2}{x^2} + \frac{3xh}{x^2} - \frac{3x^2}{x^2} = 0 \quad (\text{since } x^2 \neq 0)$$

$$2 \left(\frac{h}{x} \right)^2 + 3 \left(\frac{h}{x} \right) - 3 = 0$$

✓

$$\therefore \frac{h}{x} = \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times -3}}{2 \times 2}$$

$$= \frac{-3 \pm \sqrt{33}}{4}$$

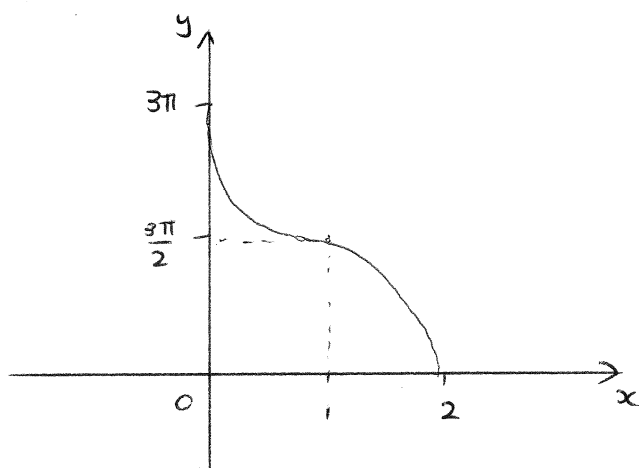
BUT $\frac{h}{x} > 0 \quad \therefore \frac{h}{x} = \frac{-3 + \sqrt{33}}{4}$

✓ (must justify positive answer only)

QUESTION 14:

a) $y = 3 \cos^{-1}(x-1)$

i)



✓ shape
✓ intercepts

ii) METHOD 1:

$$A = \int_0^{3\pi} \cos \frac{y}{3} + 1 \, dy \quad \checkmark$$

$$= \left[3 \sin \frac{y}{3} + y \right]_0^{3\pi}$$

$$= 3 \sin \frac{3\pi}{3} + 3\pi - (3 \sin 0 + 0)$$

$$= 3\pi \text{ unit}^2$$

} ✓

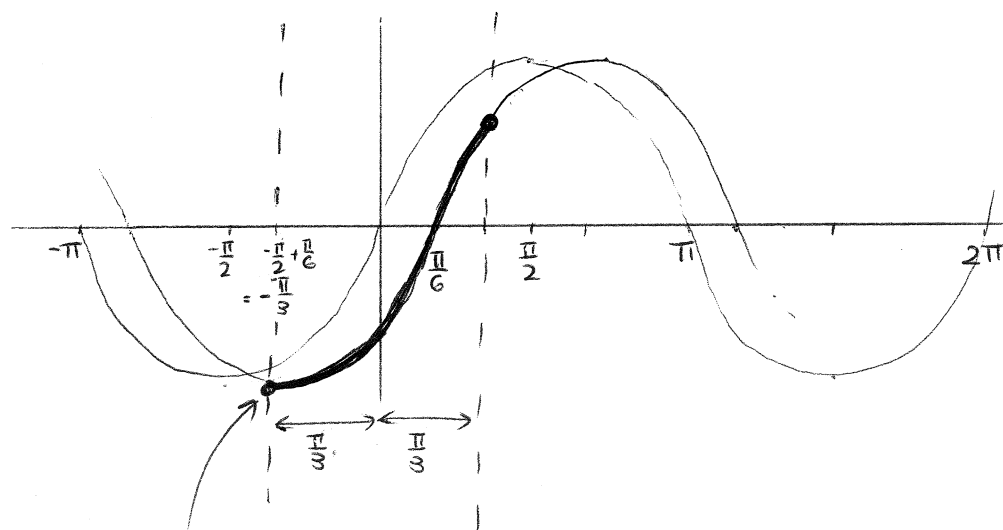
METHOD 2:

From symmetry about $(1, \frac{3\pi}{2})$: ✓

$$A = \frac{3\pi}{2} \times 2$$

$$= 3\pi \text{ unit}^2 \quad \checkmark$$

b) $f(x) = \sin(x - \frac{\pi}{6})$



$f(x) = \sin(x - \frac{\pi}{6})$
will fail the
horizontal line
test outside of
 $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$ ✓

beyond this point, the
inverse will no longer
be a function

$$-\frac{\pi}{2} + \frac{\pi}{6} = -\frac{\pi}{3}$$

$\therefore$ max possible value of $a = \frac{\pi}{3}$. ✓

c) i) $\frac{dx}{dt} = 2a$

$$\frac{dy}{dt} = -2at$$

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$= -2at \times \frac{1}{2a}$$

$$= -t$$

$$\therefore m_{\perp} = \frac{1}{t} \quad \checkmark$$

some working
required.

$$y - (-at^2) = \frac{1}{t}(x - 2at)$$

$$ty + at^3 = x - 2at$$

$$\therefore x - ty = 2at - at^3 \text{ as required.} \quad \checkmark$$

$$\text{ii) } x - ty = 2at + at^3 \quad (1)$$

$$x = -at \quad (2)$$

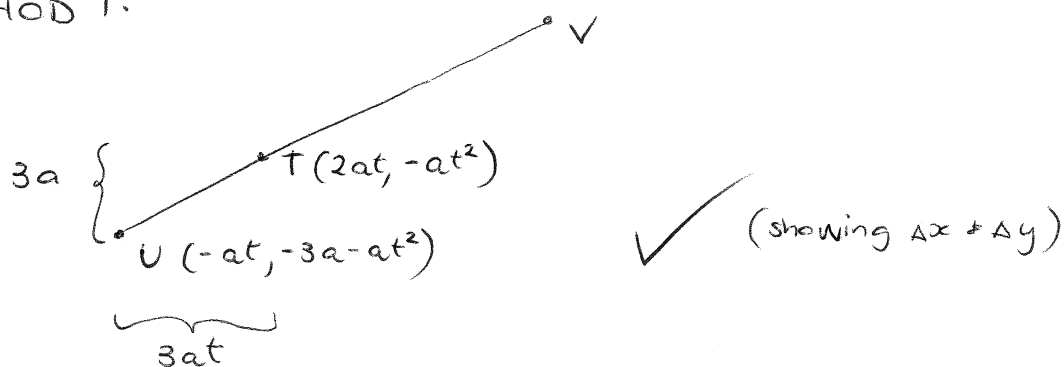
Sub (2) into (1):

$$-at - ty = 2at + at^3 \quad \checkmark$$

$$\therefore y = -3a - at^2$$

$$\therefore V(-at, -3a - at^2) \quad \checkmark$$

iii) METHOD 1:



$$V(2at + 2 \times 3at, -at^2 + 2 \times 3a)$$

$$\rightarrow V(8at, 6a - at^2) \quad \checkmark$$

METHOD 2:

$$x = \frac{3 \times 2at + (-2) \times (-at)}{-2 + 3}$$

$$= \frac{6at + 2at}{1}$$

$$= 8at \quad \checkmark$$

$$y = \frac{3 \times (-at^2) + (-2) \times (-3a - at^2)}{3 + (-2)}$$

$$= -3at^2 + 6a + 2at^2$$

$$= 6a - at^2$$

$$\therefore V(8at, 6a - at^2) \quad \checkmark$$

$$\text{iv)} \quad x = 8at \rightarrow t = \frac{x}{8a} \quad (1)$$

$$y = 6a - at^2 \quad (2)$$

Sub (1) into (2):

$$y = 6a - a\left(\frac{x}{8a}\right)^2$$

$$= 6a - \frac{x^2}{64a} \quad \checkmark$$

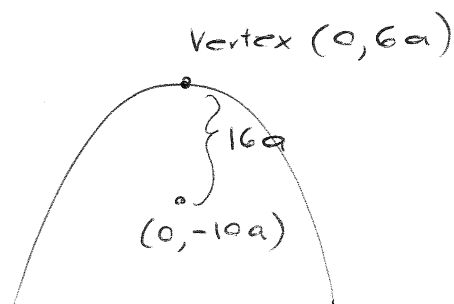
$$x^2 = (6a - y) \times 64a$$

$$= -64a(y - 6a) \quad \checkmark$$

$$4A = 64a$$

$$A = \frac{64a}{4}$$

$$= 16a$$



$\therefore$ The focus is $(0, -10a) \quad \checkmark$